

## Submission STAT1-FINAL-SUB02

### Question STAT1-FINAL-Q01

work / setup:

1.  $\bar{x} = 26/5 = 5.20$

crossed-out arithmetic: [smudged]

FINAL (maybe): mean=5.20; sample variance=2.20 plus 1

### Question STAT1-FINAL-Q02

work / setup:

1.  $P(D_1 \cap D_2) = (4/15)((4-1)/(15-1))$

crossed-out arithmetic: [smudged]

FINAL (maybe): 0.0571 plus 1

### Question STAT1-FINAL-Q03

work / setup:

1.  $SE = \sqrt{(p(1-p))/n} = \sqrt{(0.45(1-0.45)/40)} = 0.0787$

2.  $CI = 0.45 \pm 1.96(0.0787) = (0.296, 0.604)$

crossed-out arithmetic: [smudged]

FINAL (maybe): (0.296, 0.604); repeated-sampling interpretation plus 1

### Question STAT1-FINAL-Q04

work / setup:

1. Increase sample size  $n$  to reduce standard error.

2. Reduce confidence level or population variability when justified.

crossed-out arithmetic: [smudged]

FINAL (maybe): increase  $n$ ; lower confidence level or variability; note the coverage trade-off plus 1

### Question STAT1-FINAL-Q05

work / setup:

1.  $0.03 < 0.05$ , so reject  $H_0$  under the stated rule.

crossed-out arithmetic: [smudged]

FINAL (maybe): reject  $H_0$ ; evidence against  $H_0$ , not proof that  $H_0$  is false plus 1

### Question STAT1-FINAL-Q06

work / setup:

1.  $\bar{x} = 31/5 = 6.20$

crossed-out arithmetic: [smudged]

FINAL (maybe): mean=6.20; sample variance=1.70 plus 1

### Question STAT1-FINAL-Q07

work / setup:

1.  $P(D_1 \cap D_2) = (5/19)((5-1)/(19-1))$

crossed-out arithmetic: [smudged]

FINAL (maybe): 0.0585 plus 1

Question STAT1-FINAL-Q08

work / setup:

1.  $SE = \sqrt{p(1-p)/n} = \sqrt{(0.35(1-0.35)/80)} = 0.0533$

2.  $CI = 0.35 \pm 1.96(0.0533) = (0.245, 0.455)$

crossed-out arithmetic: [smudged]

FINAL (maybe): (0.245, 0.455); repeated-sampling interpretation plus 1

Question STAT1-FINAL-Q09

work / setup:

1. Increase sample size  $n$  to reduce standard error.

2. Reduce confidence level or population variability when justified.

crossed-out arithmetic: [smudged]

FINAL (maybe): increase  $n$ ; lower confidence level or variability; note the coverage trade-off plus 1

Question STAT1-FINAL-Q10

work / setup:

1.  $0.03 < 0.05$ , so reject  $H_0$  under the stated rule.

crossed-out arithmetic: [smudged]

FINAL (maybe): reject  $H_0$ ; evidence against  $H_0$ , not proof that  $H_0$  is false plus 1